3.4 Pre-Test: Trigonometry Review — Markscheme

1. [Maximum mark: 11]

   Circle centre O, $r = 8$, $A\hat{O}B = 2.1$ radians.

   (a) $l = 8 \times 2.1$ **M1**

   $= 16.8$ cm **A1**

   Wrong method: the degree formula, $\frac{2.1}{360} \times 2\pi \times 8 = 0.293$*, earns M0A0.*

   (b) $AB^2 = 8^2 + 8^2 - 2 \times 8 \times 8\cos 2.1$ **M1A1**

   $AB = 13.9$ cm $\quad (13.8787\ldots)$ **A1**

   M1 for substituting into the cosine rule, A1 for correct substitution. Accept the right-triangle method $AB = 2 \times 8\sin 1.05$ *for M1A1. Wrong method: degree mode gives* $AB = 0.293$ *(0.293198…), earning M1A1A0.* $AB^2 = 192.620\ldots$ *given as the answer earns M1A1A0.*

   (c) area of sector $= \dfrac{1}{2} \times 8^2 \times 2.1 = 67.2$ **A1**

   area of triangle $AOB = \dfrac{1}{2} \times 8 \times 8\sin 2.1 = 27.6226\ldots$ **A1**

   shaded area $= 67.2 - 27.6226\ldots$ **M1**

   $= 39.6$ cm^2 $\quad (39.5773\ldots)$ **A1**

   M1 for subtracting the triangle from the sector (their values). Accept 39.6 from the rounded values $67.2-27.6$*. Wrong methods: degree mode for the triangle,* $\frac{1}{2}\times 64\sin 2.1^\circ = 1.17$*, gives* 66.0*, earning A1A0M1A0; the sector area alone (67.2) earns A1A0M0A0; r for* r^2 *in the sector,* $\frac{1}{2} \times 8 \times 2.1 = 8.4$*, earns A0.*

   (d) perimeter $= 16.8 + 13.8787\ldots$ **M1**

   $= 30.7$ cm $\quad (30.6787\ldots)$ **A1**

   M1 for arc plus chord. FT from (a) and (b). Wrong method: adding the two radii, $16.8 + 13.88 + 16 = 46.7$*, or the sector perimeter* $16.8 + 16 = 32.8$*, earns M0A0.*

2. [Maximum mark: 12]

Garden: $AB = 45$, $BC = 60$, $AC = 80$.

(a) The largest angle is $A\hat{B}C$, opposite the longest side AC.

$\cos A\hat{B}C = \dfrac{45^2 + 60^2 - 80^2}{2 \times 45 \times 60}$ **M1A1**

$A\hat{B}C = 98.3°$ $(98.2514\ldots)$ **A1**

M1 for substituting into the cosine rule for any angle, A1 for the angle opposite 80. $\cos A\hat{B}C = -0.143518\ldots$ *(*$-\frac{775}{5400}$*). Wrong methods: the angle opposite* 45, $33.8°$*, or opposite* 60, $47.9°$*, earns M1A0A0; radian mode gives* 1.71 *(1.71481…): accept it only with the unit "radians" stated, A1A0 otherwise.*

(b) area $= \dfrac{1}{2} \times 45 \times 60 \sin 98.2514\ldots°$ **M1**

$= 1340$ m^2 $(1336.02\ldots)$ **A1**

Accept any correct pairing of two sides with their included angle. Accept 1336. FT from (a). Wrong method: an angle that is not included between the two sides, e.g. $\frac{1}{2} \times 45 \times 60 \sin 33.8°$*, earns M0A0.*

Triangle PQR: $PQ = 14$, $QR = 10$, $Q\hat{P}R = 38°$.

(c) $\dfrac{\sin P\hat{R}Q}{14} = \dfrac{\sin 38°}{10}$ **M1**

$\sin P\hat{R}Q = 0.861926\ldots$

$P\hat{R}Q = 59.5°$ $(59.5335\ldots)$, $P\hat{R}Q = 120°$ $(120.466\ldots)$ **A1A1**

Accept $120.5°$*. Wrong methods: only the acute angle earns M1A1A0; sides paired with the wrong angles,* $\sin P\hat{R}Q = \frac{10 \sin 38°}{14}$*, gives* $26.1°$*, M0A0A0.*

(d) Method 1 (sine rule).

$P\hat{Q}R = 180 - 38 - 59.5335\ldots = 82.4665\ldots°$ or $P\hat{Q}R = 180 - 38 - 120.466\ldots = 21.5335\ldots°$ **M1**

$PR = \dfrac{10 \sin 82.4665\ldots°}{\sin 38°} = 16.1$ $(16.1024\ldots)$ **A1**

$PR = \dfrac{10 \sin 21.5335\ldots°}{\sin 38°} = 5.96$ $(5.96181\ldots)$ **A1**

difference $= 16.1024\ldots - 5.96181\ldots = 10.1$ cm $(10.1406\ldots)$ **A1**

M1 for a third angle found from either value in (c). Method 2, cosine rule: $10^2 = 14^2 + PR^2 - 2 \times 14 \times PR \cos 38°$ *(M1), so* $PR^2 - 22.0643\ldots PR + 96 = 0$*,* $PR = 16.1$ *or* 5.96 *(A1A1), difference* 10.1 *(A1). Accept* $16.1 - 5.96 = 10.14$*. FT from (c). Wrong methods: one length only earns M1A1A0A0; the sum* 22.1 *earns the final A0.*

3. [Maximum mark: 9]

(a)(i) $\sin\dfrac{4\pi}{3} = -\dfrac{\sqrt{3}}{2}$ **A1**

(a)(ii) $\tan\dfrac{2\pi}{3} = -\sqrt{3}$ **A1**

Accept $-\frac{3}{\sqrt{3}}$. *Decimals from the GDC* (-0.866, -1.73) *earn A0: exact values are asked for. Common errors, each A0: the wrong sign;* $\cos\frac{4\pi}{3} = -\frac{1}{2}$ *given for (i);* $-\frac{1}{\sqrt{3}}$ *(ratio inverted) for (ii).*

(b)(i) $\cos^2\theta = 1 - \left(\dfrac{2}{3}\right)^2 = \dfrac{5}{9}$ **M1**

$\cos\theta = \pm\dfrac{\sqrt{5}}{3}$ **A1**

$\cos\theta = -\dfrac{\sqrt{5}}{3}$, since θ is in the second quadrant **A1**

(b)(ii) $\tan\theta = \dfrac{2/3}{-\sqrt{5}/3} = -\dfrac{2}{\sqrt{5}}$ **A1**

Accept a right-triangle method (sides 2, $\sqrt{5}$*, 3) for the M1A1. The final A1 in (i) is for the negative sign. Accept* $-\frac{2\sqrt{5}}{5}$ *in (ii). Wrong method: the GDC route* $\cos\left(\arcsin\frac{2}{3}\right) = +\frac{\sqrt{5}}{3}$ *gives the first-quadrant angle, earning M1A1A0, and then* $\tan\theta = +\frac{2}{\sqrt{5}}$*, A0 unless FT. FT (ii) from the candidate's* $\cos\theta$*. Inverted ratio* $\frac{\cos\theta}{\sin\theta} = -\frac{\sqrt{5}}{2}$ *earns A0. Decimals* (-0.745, -0.894) *earn A0.*

(c) $\cos x = -\dfrac{\sqrt{3}}{2}$ **M1**

$x = \dfrac{5\pi}{6},\ \ x = \dfrac{7\pi}{6}$ **A1A1**

Wrong methods: the GDC gives only $\frac{5\pi}{6}$ *(2.618), earning M1A1A0;* $x = \frac{\pi}{6}, \frac{11\pi}{6}$ *(sign of* $\cos x$ *wrong) earns M0A0A0;* $\frac{5\pi}{6}$ *and* $-\frac{5\pi}{6}$ *(*$-\frac{5\pi}{6}$ *outside the interval) earns M1A1A0. Decimals* (2.62, 3.67) *earn A1A0 at most: the question asks for multiples of* π*. Extra values outside the interval earn the final A0.*

4. [Maximum mark: 13]

$h(t) = d - a\cos(bt)$; lowest 2 m, highest 30 m, period 12 minutes.

(a) $a = \dfrac{30-2}{2} = 14$ **A1**

$d = \dfrac{30+2}{2} = 16$ **A1**

Wrong methods, each A0: $a = 28$ *(not halved);* $d = 14$.

(b) $b = \dfrac{2\pi}{12}$ **M1**

$b = \dfrac{\pi}{6}$ $(0.523598\ldots)$ **A1**

Accept 0.524. *Wrong methods:* $b = 12$ *(the period used as b), or* $b = 2\pi \times 12 = 75.4$, *earns M0A0;* $b = 30$ *(degree measure) earns M1A0.*

(c) $h(2.5) = 16 - 14\cos\left(\dfrac{\pi}{6} \times 2.5\right)$ **M1**

$= 12.4$ m $(12.3765\ldots)$ **A1**

FT from (a), (b). $b = 30$ *in degree mode is consistent: M1A1. Wrong method:* $b = \frac{\pi}{6}$ *in degree mode gives* 2.00, *M1A0.*

(d) $16 - 14\cos\left(\dfrac{\pi}{6}t\right) = 25$ **M1**

GDC: graph $y = h(t)$ *and* $y = 25$ *for* $0 \leq t \leq 12$*; find both intersections.*

$t = 4.33$ $(4.33350\ldots)$ and $t = 7.67$ $(7.66649\ldots)$ **A1**

length $= 7.66649\ldots - 4.33350\ldots = 3.33$ minutes $(3.33298\ldots)$ **A1**

M1 for $h(t) = 25$, $h(t) > 25$, *or a sketch with the line* $h = 25$. *The first A1 needs both values. Accept 3.34 from* $7.67 - 4.33$. *Wrong methods: one intersection (4.33) as the answer, M1A0A0; the time below 25 m,* 8.67, *M1A1A0. FT from the candidate's h.*

(e) Reflection in the t-axis (horizontal axis) **A1**

Vertical stretch, scale factor 14 **A1**

Horizontal stretch, scale factor $\dfrac{6}{\pi}$ $(1.90985\ldots)$ **A1**

Vertical translation 16 up **A1**

"Describe fully" needs words (markscheme-spec §19): one mark per transformation, with its type, direction and size. Accept the scale factor 1.91 *and "translation by* $\binom{0}{16}$*". Reject "move", and "flip" with no axis. Vertical stretch with scale factor* -14 *earns the stretch mark, not the reflection mark. Equations with no words earn at most 1 mark. Order: reflection and vertical stretch (either order) before the vertical translation; deduct 1 mark for a wrong order. Wrong method: horizontal scale factor* $\frac{\pi}{6}$, *A0. FT from (a) and (b).*